

Meeks, Knight, Brewer, Cook 2014 - Marsh

Investigating the subjective reports of rejection processes in the word frequency mirror effect

Auto-assembled report. Section content is drawn from the summarization pipeline outputs.

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Paper Synthesis

Paper: Meeks, Knight, Brewer, Cook 2014 - Marsh Investigating the subjective reports of rejection processes in the word frequency mirror effect

Executive Summary

The paper investigates whether the lower false-alarm rate for low-frequency words stems from a metacognitive memorability judgment or a lack of familiarity, using subjective rejection classifications across three experiments. It finds that low-frequency items are more often rejected on the basis of perceived memorability in mixed-list contexts, but this effect disappears

in pure-list designs, while the basic correct-rejection advantage persists. These results suggest that memorability-based rejection contributes to the word-frequency mirror effect only when contrasting frequency items are present, highlighting the role of test context in metacognitive judgments.

Reading Recommendation

Read Selectively

The paper offers novel insights into how word-frequency context influences memorability-based rejection judgments, but methodological gaps (no power analysis, incomplete ANOVA reporting, uncorrected multiple comparisons) and overextension to distinctiveness without measurement limit its rigor. Experts should focus on the experimental methods, key results (especially the interaction of list composition with memorability judgments), and the authors' theoretical discussion.

Background and Motivation

The paper addresses the unresolved question of whether the lower false-alarm advantage for low-frequency words stems from a metacognitive memorability assessment or from a lack of pre-experimental familiarity, a gap left open by earlier mirror-effect research. This gap is motivated by classic findings of the word-frequency mirror effect (Glanzer & Adams, 1985) and by evidence that low-frequency items are more memorable (Benjamin, 2003) yet less familiar (Underwood & Freund, 1970). Recent work showing independent mirror effects of word frequency and context variability (Cook et al., 2006) further underscores the need to disentangle these competing processes.

Theoretical Stance

Overall stance: Consistent Theoretical

The introduction foregrounds a memorability-based rejection account and the discussion repeatedly returns to this account as the primary explanation, while treating alternative accounts as competing or supplementary explanations without abandoning the focal theory.

Theories Discussed

- **memorability-based rejection process** (*focal*): A metacognitive judgment that low-frequency items are more memorable, leading participants to reject them as new because they would have been remembered if studied.
- **familiarity-based rejection** (*alternative*): Rejection of low-frequency items due to their lower pre-experimental familiarity, producing weaker implicit associative responses.

- **distinctiveness heuristic** (*alternative*): Use of an item's distinctiveness (e.g., diagnostic details) as evidence for non-studied status; low-frequency items are inherently more distinctive.
- **inherent/secondary distinctiveness account** (*background*): The notion that the infrequent occurrence of low-frequency words makes them intrinsically more distinctive and thus more memorable.
- **recall-to-reject mechanism** (*background*): A process whereby participants retrieve contextual information from study to determine that a test item was not previously encountered.
- **functional account of recollection rejection** (*background*): A view that recollection-based rejection depends on contextual factors and criterial status of recollection judgments.

Study Design

- **Number of experiments:** 3
- **Predictors examined:** word frequency, context variability, consonant-vowel status
- **Outcomes examined:** hit rate, remember responses, know responses, guess responses, correct rejection rate, would have remembered responses, lacks familiarity responses, recall to reject responses
- **Materials:** Nouns drawn from the Kucera-Francis (1967) norms (Exp 1) and from the Steyvers & Malmberg (2003) corpus (Exp 2-3), controlled for length, syllables, concreteness, and imagery; low-frequency words (~10 per million) and high-frequency words (~88 per million) were combined with low/high context variability or consonant-vowel status manipulations.
- **Design types:**
 - exp_1: within-subjects 1-factor (low vs. high word frequency)
 - exp_2: within-subjects 2×2 factorial (word frequency $\times$ context variability)
 - exp_3: mixed design - between-subjects word frequency $\times$ within-subjects consonant-vowel status

Findings Summary

Across the three experiments, low-frequency words (LWF) consistently yielded higher correct-rejection rates than high-frequency words (HWF), but the subjective basis for this advantage differed by design. In mixed-list (within-subjects) experiments 1 and 2, participants

were more likely to attribute LWF rejections to a "Would Have Remembered" (memorability-based) process, whereas HWF rejections were linked to "Lacks Familiarity"; the recall-to-reject option showed no frequency effect. Experiment 3, which used pure-list (between-subjects) frequency manipulation, replicated the LWF correct-rejection advantage but eliminated the memorability-based classification difference, with no significant differences in "Would Have Remembered" or "Lacks Familiarity" responses between LWF and HWF items. Context variability produced parallel, independent effects on hits, remembers, knows, and subjective classifications, confirming that WF and CV each drive mirror-effect patterns. Thus, the memorability-based rejection advantage for LWF items appears contingent on the presence of contrasting frequency items in the test context, while the basic correct-rejection benefit persists across list compositions.

Evidence-Claim Alignment

Supported Claims

- Low-frequency (LWF) words produce higher correct-rejection rates than high-frequency (HWF) words - supported by Experiment 1 $t(24)=4.959$, $p<.001$ and replicated in Experiments 2 and 3 (significant main effects of WF on false-alarm rates).
- In mixed-list designs (Experiments 1 and 2), participants attribute LWF correct rejections more often to a "Would Have Remembered" (memorability-based) response than HWF items - supported by Experiment 1 $t(24)=4.381$, $p<.001$ and Experiment 2 main effects of WF on Would-Have-Remembered responses ($F(1,33)=15.808$, $p<.001$).
- HWF correct rejections are more often classified as "Lacks Familiarity" - supported by Experiment 1 conditionalized Lacks-Familiarity responses ($t(24)=4.186$, $p<.001$) and Experiment 2 main effect of CV on Lacks-Familiarity ($F(1,33)=9.059$, $p=.005$).
- Word frequency (WF) and context variability (CV) each produce independent mirror-effect patterns on hits, remembers, knows, and subjective classifications - supported by Experiment 2 main effects of WF and CV on hit rates, Remember, Know, and Would-Have-Remembered responses (all $p\leq.001$).
- In pure-list (between-subjects) designs, the LWF correct-rejection advantage persists even though memorability-based classifications do not differ - supported by Experiment 3 main effect of WF on correct-rejection rates ($F(1,67)=4.066$, $p=.048$) and null WF effects on Would-Have-Remembered and Lacks-Familiarity responses (both $p>.4$).

Unsupported or Weakly Supported Claims

- The claim that memorability-driven rejection fully accounts for the LWF correct-rejection advantage - evidence is weak because alternative processes (Lacks-Familiarity,

Recall-to-Reject) showed non-significant differences ($p > .12$) and were not integrated into the theoretical narrative.

- The assertion that distinctiveness (inherent or secondary) explains the effect - no direct measure of item distinctiveness was collected, so the claim extends beyond the presented data.
- The prediction that participants would show no difference in hit rates for LWF versus HWF items in pure-list designs - Experiment 3 found no significant WF effect on hits ($p = .283$), but the authors still discuss a "failure to find some of these effects" without providing a definitive conclusion, leaving the claim ambiguous.

Contradictions or Tensions

- Experiments 1 and 2 show a strong memorability-based classification advantage for LWF items, yet Experiment 3 (pure-list) eliminates this classification difference while the correct-rejection advantage remains, creating tension with the overarching claim that memorability drives the rejection benefit.
- The discussion generalizes the findings to a distinctiveness heuristic despite the absence of distinctiveness metrics, contradicting the earlier emphasis on memorability as the primary mechanism.
- Authors claim a "mirror effect" for both WF and CV, but the marginal CV effect on correct-rejection rates in Experiment 2 ($p = .058$) is presented as supportive, which conflicts with the standard significance threshold and weakens the claim of independent CV mirror effects.

Strengths

- Clear experimental progression across three studies that isolates word frequency, context variability, and list composition effects on recognition memory.
- Use of novel subjective classification categories (Would Have Remembered, Lacks Familiarity, Recall-to-Reject) to directly probe participants' metacognitive rejection processes.
- Consistent replication of the low-frequency correct-rejection advantage across mixed-list and pure-list designs, providing converging evidence for the mirror effect.
- Demonstration that word frequency and context variability each produce independent main effects on hits, Remember, Know, and subjective classifications, supporting the notion of multiple contributing mechanisms.

- Inclusion of conditionalized response analyses that control for overall response tendencies, strengthening the interpretation of frequency-specific classification patterns.
- Transparent reporting of effect sizes (Cohen's d , η^2 , g^2) for significant findings, facilitating assessment of practical significance.

Weaknesses and Concerns

- Theoretical overreach: the authors extend their findings to a distinctiveness account without measuring item distinctiveness, weakening the claim that distinctiveness explains the memorability advantage.
- Alternative processes (Lacks Familiarity, Recall-to-Reject) showed non-significant effects yet are downplayed; the narrative that memorability fully accounts for the rejection advantage is not fully supported by the data.
- Statistical reporting gaps: the main effect of Response Type in the 2×3 ANOVA (Experiment 1) is omitted, and multiple pairwise t-tests are presented without correction for family-wise error, raising concerns about Type I inflation.
- Methodological omissions: no a priori power analyses, absent counterbalancing descriptions, and insufficient exclusion criteria across all experiments limit confidence in the robustness of the results.
- Reliability of the subjective classification measures is not assessed, leaving unclear whether the reported response categories consistently capture the intended constructs.
- Marginal findings (e.g., context variability effect on correct rejections in Experiment 2, $p = .058$) are treated as supportive evidence, which may overstate the strength of the independent CV mirror effect.

Key References

- **Glanzer & Adams (1985)** -- foundational theory
- **Benjamin (2003)** -- primary methodological precedent
- **Underwood & Freund (1970)** -- competing theoretical account
- **Israel & Schacter (1997)** -- competing theoretical account
- **Cook et al. (2006)** -- key empirical precedent
- **Steyvers & Malmberg (2003)** -- key empirical precedent
- **Ghetti (2003)** -- primary methodological precedent

Abstract

Preserved from source without summarization.

We sought to systematically investigate how participants subjectively classify the basis of their recognition memory judgments for low and high word frequency items. We found that participants more often reported rejection processes related to the increased perceived memorability for unstudied low word frequency items (relative to high word frequency items), rather than classifying their decision on a lack of familiarity. Experiment 2 replicated this pattern and demonstrated context variability and word frequency independently influenced the subjective classifications for correct rejections. Results of Experiment 3 revealed that these differences are dependent upon having experience with both low and high frequency items. Overall, these data suggest participants' rejection of low frequency items is more strongly related to judgments of perceived memorability, but only when they are presented in the context of high frequency items. The results are discussed in relation to distinctiveness and expected memorability.

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Introduction (Paper-Level)

Section: paper-level

Research Gap / Motivation

It remains unknown whether participants rely primarily on a metacognitive memorability assessment or on a lack of familiarity to correctly reject low-frequency items.

Theoretical Framework

memorability-based rejection process

This metacognitive, memorability-based judgment requires that the participant realize that LWF items are more memorable than HWF items.

- Citations: Benjamin (2003); Brown, Lewis, & Monk (1977); Guttentag & Carroll (1998)

familiarity-based rejection process

Because LWF items lack pre-experimental familiarity relative to HWF items, they elicit fewer implicit associative responses (Underwood & Freund, 1970). This lower level of familiarity may result in greater rejections of new LWF items as studied in the experimental context.

- Citations: Underwood & Freund (1970)

distinctiveness heuristic similarity

The idea of a memorability-based rejection process bears similarity to research on the distinctiveness heuristic.

- Citations: Israel & Schacter (1997)

Specific Hypotheses / Predictions

memorability vs familiarity hypothesis

Our primary goal in Experiment 1 was to assess whether the increased correct rejection rate for LWF items is related to a subjective judgment of increased memorability or to a lack of familiarity.

context variability confound hypothesis

We also sought to determine whether the rejection processes associated with the increased rejection of LWF items are confounded with the rejection processes for a word characteristic that is highly correlated with WF (i.e., context variability; Experiment 2).

- Citations: Steyvers & Malmberg (2003)

pure list WF hypothesis

In Experiment 3, we examined whether the differences in rejection strategies would still apply in pure lists (i.e., WF manipulated between-subjects).

Mechanistic Claims

memorability-based rejection mechanism

This metacognitive, memorability-based judgment requires that the participant realize that LWF items are more memorable than HWF items.

- Citations: Benjamin (2003)

familiarity lack rejection mechanism

LWF items lack pre-experimental familiarity relative to HWF items, they elicit fewer implicit associative responses... This lower level of familiarity may result in greater rejections of new LWF items.

- Citations: Underwood & Freund (1970)

recall-to-reject mechanism

We also included a recall-to-reject option in Experiment 1 as it has traditionally been examined in studies on rejection processes (e.g., Ghetti, 2003).

- Citations: Ghetti (2003)

Constructs of Interest

- **word frequency (low vs high) (*predictor*)**: Normative frequency of a word in the language; low-frequency words occur less often in everyday exposure.
- **correct rejection rate (false alarm rate) (*outcome*)**: Proportion of new items correctly identified as new; lower false alarms indicate higher correct rejection.

Rationale for Predictor

Low-frequency items have fewer pre-experimental associations and may be perceived as more distinctive or less familiar, potentially influencing rejection decisions.

Rationale for Outcome

Understanding the processes that lead to lower false alarm rates for low-frequency items clarifies how rejection mechanisms contribute to the word-frequency mirror effect.

Key Terminology

memorability-based rejection process: A metacognitive judgment that low-frequency items are more memorable, leading participants to reject them as new because they would have been remembered if studied.

This metacognitive, memorability-based judgment requires that the participant realize that LWF items are more memorable than HWF items.

distinctiveness heuristic: A metacognitive strategy where the absence of distinctive, diagnostic details leads participants to infer that an item was not studied.

This result presumably occurs because participants metacognitively use the absence of diagnostic, pictorial details in memory as evidence that if they had studied the item, they would have remembered it.

recall-to-reject process: A rejection mechanism in which participants retrieve contextual information from study to determine that a test item was not previously encountered.

a "do not recall to reject" process; see Strack & Bless, 1994 for a similar idea.

Background Research

- Strong (1912) highlighted the long history of context-based item detection.
- Glanzer et al. (1993) reviewed mirror effects in recognition memory.
- Glanzer & Adams (1985) documented the normative word-frequency mirror effect.
- Arndt & Reder (2002) and others linked false alarms to pre-experimental familiarity of high-frequency words.
- Cook et al. (2006) and related work showed more Remember responses for low-frequency items.
- Donaldson (1996) and subsequent critiques questioned the confidence interpretation of Remember/Know judgments.
- Underwood & Freund (1970) reported fewer implicit associative responses for low-frequency items.
- Israel & Schacter (1997) demonstrated the distinctiveness heuristic reducing false memory.

Alternative Theories Discussed

familiarity-based rejection

- **Description:** Proposes that low-frequency items are rejected because they lack pre-experimental familiarity, leading to weaker implicit associative responses.
- **Difference from focal theory:** Focuses on absence of familiarity rather than a metacognitive memorability judgment.
- **Citations:** Underwood & Freund (1970)

distinctiveness heuristic

- **Description:** Suggests that participants use the distinctiveness of an item's encoding (e.g., pictorial details) as evidence for rejection.
- **Difference from focal theory:** Relies on diagnostic detail absence rather than word-frequency-based memorability.
- **Citations:** Israel & Schacter (1997)

Overview of Studies

Experiment 1 examined memorability versus familiarity judgments for LWF and HWF new items, including a recall-to-reject option. Experiment 2 tested whether these rejection processes are confounded with context variability. Experiment 3 evaluated the same processes in pure lists with between-subjects WF manipulation.

Missing or Unclear

- Explicit operational definitions of "memorability" and "familiarity" are not provided.
- Statistical analysis plan for the hypotheses is not described in the introduction.

All Citations Referenced

- **Strong (1912)** -- history of context-based item detection
- **Malmberg (2008)** -- review of mirror effects
- **Yonelinas (2002)** -- review of mirror effects
- **Glanzer, Adams, Iverson, & Kim (1993)** -- review of mirror effects
- **Glanzer & Adams (1985)** -- normative word-frequency mirror effect
- **Arndt & Reder (2002)** -- pre-experimental familiarity
- **Guttentag & Carroll (1994)** -- pre-experimental familiarity
- **Joordens & Hockley (2000)** -- pre-experimental familiarity
- **Reder et al. (2000)** -- pre-experimental familiarity
- **Rugg, Cox, Doyle, & Wells (1995)** -- pre-experimental familiarity
- **Cook, Marsh, & Hicks (2006)** -- Remember responses for LWF
- **Dewhurst, Brandt, & Sharp (2004)** -- Remember responses
- **Dewhurst, Hitch, & Barry (1998)** -- Remember responses
- **Gardiner & Java (1990)** -- Remember responses
- **Donaldson (1996)** -- confidence interpretation of Remember/Know
- **Dunn (2004)** -- confidence interpretation
- **Wais, Mickes, & Wixted (2008)** -- confidence interpretation
- **Williams, Conway, & Moulin (2013)** -- evidence against confidence view
- **McCabe & Balota (2007)** -- list composition effects
- **Benjamin (2005)** -- test composition effects
- **Bodner & Lindsay (2003)** -- test composition effects
- **Tousignant & Bodner (2012)** -- test composition effects

- **Migo, Mayes, & Montaldi (2012)** -- review of Remember-Know
- **Lampinen, Neuschatz, & Payne (1997)** -- pros and cons of subjective reports
- **Underwood & Freund (1970)** -- implicit associative responses
- **Benjamin (2003)** -- memorability judgments
- **Brown, Lewis, & Monk (1977)** -- memorability judgments
- **Guttentag & Carroll (1998)** -- memorability judgments
- **Wixted (1992)** -- high-frequency memorability
- **Israel & Schacter (1997)** -- distinctiveness heuristic
- **Ghetti (2003)** -- do not recall to reject
- **Strack & Bless (1994)** -- similar idea
- **Rotello & Heit (2000)** -- recall-to-reject
- **Brainerd, Reyna, & Estrada (2006)** -- recollection rejection reduces false memory
- **Gallo (2004)** -- recollection rejection
- **Schmid, Herholz, Brandt, & Buchner (2010)** -- recollection rejection
- **Light (2012)** -- factors influencing recollection rejection
- **Lampinen, Arnal, & Hicks (2009)** -- instructions to use rejection process
- **Rotello, Macmillan, & Van Tassel (2000)** -- instructions to use rejection process
- **Steyvers & Malmberg (2003)** -- context variability

Additional Quoted Evidence

This metacognitive, memorability-based judgment requires that the participant realize that LWF items are more memorable than HWF items.

Because LWF items lack pre-experimental familiarity relative to HWF items, they elicit fewer implicit associative responses (Underwood & Freund, 1970).

The idea of a memorability-based rejection process bears similarity to research on the distinctiveness heuristic.

Experiment 1 -- Introduction

Not available for this paper.

Experiment 1 -- Methods

Section: exp_1

Participants

- **N:** 25
- **Recruitment:** University of Georgia undergraduate students
- **Compensation:** course credit
- **Testing context:** tested individually in a session lasting approximately 30 min

Design

Design statement: 1 (word frequency: low vs. high) within-subjects design.

Predictors:

- **word frequency**
- Levels: low (LWF), high (HWF)
- Manipulation: within-subjects
- Factor type: nominal

Counterbalancing: not reported

Outcome Measures

- **hit rate:** proportion of studied items correctly identified as old
- **remember responses:** proportion of hits classified as 'Remember'
- **know responses:** proportion of hits classified as 'Know'
- **guess responses:** proportion of hits classified as 'Guess'
- **correct rejection rate:** proportion of distractor items correctly identified as new
- **would have remembered responses:** proportion of new items classified as 'Would Have Remembered'
- **lacks familiarity responses:** proportion of new items classified as 'Lacks Familiarity'
- **recall to reject responses:** proportion of new items classified as 'Recall to Reject'

Materials

- **Description:** 120 nouns selected from Kucera and Francis (1967) word norms, controlled for length, syllables, concreteness, and imagery

- **Source:** Kucera and Francis (1967) norms
- **Stimulus pool size:** 120
- **Stimuli presented per participant:** 120 (all words presented to each participant)
- **Additional details:**
 - 60 low-frequency words (M = 10 per million)
 - 60 high-frequency words (M = 88.1 per million)

Procedure

Ordered steps:

1. Participants received written and verbal instructions about studying words for a future memory test.
 2. Study phase: computer randomly selected 60 words (30 LWF, 30 HWF) presented sequentially for 2 s each.
 3. Experimenter reiterated instructions and confirmed participant understanding.
 4. Test phase: computer presented all 120 words (60 studied, 60 distractors; distractors half LWF, half HWF) in random order.
 5. Participants classified each item as Old/New and provided subjective judgments (Remember, Know, Guess, etc.).
- **Apparatus:** computer software (unspecified) for stimulus presentation and response collection
 - **Experimenter presence:** present
 - **Presentation format:** random selection/presentation
 - **Stimulus timing:** study words presented for 2 s each; test timing not specified (self-paced)
 - **Session duration:** approximately 30 min

Preregistration and Analysis Plan

- **Preregistration:** not reported

Author-Provided Rationale for Methodological Choices

- Test instructions based on Migo et al., 2012
- Guess response included to equate response options (Hicks & Marsh, 1999)

- Instructions similar to past experiments (Marsh et al., 2009)

Missing or Unclear

- No power analysis reported
- Exclusion criteria not stated
- Exact software/hardware details not provided
- Response key mappings not listed
- Timing details for test phase responses not reported
- Data analysis plan not described

Citations Referenced in This Section

- **Kucera and Francis (1967)** -- source of word norms and stimulus selection
- **Migo et al. (2012)** -- suggested inclusion of test instructions
- **Hicks & Marsh (1999)** -- rationale for including Guess response
- **Marsh et al. (2009)** -- comparison to past experiment instructions

Quoted Evidence from Source

Twenty-five undergraduate students from the University of Georgia participated in exchange for course credit.

Sixty were LWF items ($M = 10$ per million; range 9-11 per million) and 60 were HWF ($M = 88.1$ per million; range from 80 to 100 per million).

These words were presented for 2 s each.

Experiment 1 -- Results and Discussion

Section: exp_1

Statistical Approach

A conventional alpha level of 5% was adopted for all statistical tests across all experiments.

Key Findings

Overall hit rate was higher for LWF items than HWF items

- **Test:** independent samples t-test
- **Effect type:** main effect
- **Variables in model:** Word Frequency, Hit Rate

Verbatim statistical report:

| $t(24) = 2.357, p = .027, d = .472$

Interaction between Word Frequency and Response Type was significant

- **Test:** 2 (WF) - 3 (Response Type) repeated measures ANOVA
- **Effect type:** interaction
- **Variables in model:** Word Frequency, Response Type

Verbatim statistical report:

| $F(2, 48) = 4.208, p = .021, g^2 = .149$

Proportion of Remember responses was higher for LWF items than HWF items

- **Test:** independent samples t-test
- **Effect type:** simple effect
- **Variables in model:** Word Frequency, Remember Responses

Verbatim statistical report:

| $t(24) = 2.615, p = .015, d = .526$

Proportion of Guess responses was higher for HWF items than LWF items

- **Test:** independent samples t-test
- **Effect type:** simple effect
- **Variables in model:** Word Frequency, Guess Responses

Verbatim statistical report:

| $t(24) = 2.477, p = .021, d = .497$

Proportion of Know responses did not differ across high and low WF items

- **Test:** independent samples t-test
- **Effect type:** simple effect

- **Variables in model:** Word Frequency, Know Responses

Verbatim statistical report:

| $t(24) = 1.234, p = .229$

Correct rejection rate was higher for LWF items than HWF items

- **Test:** independent samples t-test
- **Effect type:** main effect
- **Variables in model:** Word Frequency, Correct Rejection Rate

Verbatim statistical report:

| $t(24) = 4.959, p < .001, d = .995$

Interaction between Word Frequency and Response Type for correct rejections was significant

- **Test:** 2 (WF) - 3 (Response Type) repeated measures ANOVA
- **Effect type:** interaction
- **Variables in model:** Word Frequency, Response Type

Verbatim statistical report:

| $F(2, 48) = 6.138, p = .004, \eta^2 = .204$

Proportion of Would Have Remembered responses was higher for LWF items than HWF items

- **Test:** independent samples t-test
- **Effect type:** simple effect
- **Variables in model:** Word Frequency, Would Have Remembered Responses

Verbatim statistical report:

| $t(24) = 4.381, p < .001, d = .879$

Proportion of Lacks Familiarity responses did not differ across LWF and HWF items

- **Test:** independent samples t-test
- **Effect type:** simple effect

- **Variables in model:** Word Frequency, Lacks Familiarity Responses

Verbatim statistical report:

| $t(24) = 1.210, p = .238$

Proportion of Recall to Reject responses did not differ across LWF and HWF items

- **Test:** independent samples t-test
- **Effect type:** simple effect
- **Variables in model:** Word Frequency, Recall to Reject Responses

Verbatim statistical report:

| $t(24) = 1.588, p = .125$

More conditionalized Would Have Remembered responses for LWF than HWF items

- **Test:** independent samples t-test
- **Effect type:** simple effect
- **Variables in model:** Word Frequency, Conditionalized Would Have Remembered Responses

Verbatim statistical report:

| $t(24) = 3.889, p = .001, d = .778$

More conditionalized Lacks Familiarity responses for HWF items than LWF items

- **Test:** independent samples t-test
- **Effect type:** simple effect
- **Variables in model:** Word Frequency, Conditionalized Lacks Familiarity Responses

Verbatim statistical report:

| $t(24) = 4.186, p < .001, d = .836$

Conditionalized Recall to Reject responses did not differ across LWF and HWF items

- **Test:** independent samples t-test

- **Effect type:** simple effect
- **Variables in model:** Word Frequency, Conditionalized Recall to Reject Responses

Verbatim statistical report:

$t(24) = .758, p = .456$

Authors' Interpretations

Remember responses for LWF items

With regards to reported rejection processes, LWF items were given more Would Have Remembered responses than HWF items, presumably due to their increased sense of memorability.

Lacks Familiarity responses for HWF items

In contrast, HWF items were given more conditionalized Lacks Familiarity responses than LWF items. Thus, when HWF items were correctly rejected, the decision was more likely to be related to an assessment of familiarity and not a perceived memorability process.

Recollection rejection levels

Recollection rejection levels were low overall with no difference between the two classes of WF. This finding is consistent with the idea that there have to be very specific circumstances present in order to elicit differences in recollection rejection (e.g., Brainerd et al., 2006; Gallo, 2004; Light, 2012; Schmid et al., 2010).

- Citations: Brainerd et al. (2006); Gallo (2004); Light (2012); Schmid et al. (2010)

WF mirror effect components

The results from Experiment 1 suggest that the WF mirror effect is not solely driven by the decision processes acting on memory for old items that are judged in the recognition test. An additional, important component of the WF mirror effect is found in participants' subjective classification associated with the rejection of new items that also differ in their normative WF.

Authors' Stated Links to Their Own Predictions

WF mirror effect replication

We replicated the classic WF mirror effect in the hit and correct rejection rates and the finding that there are more Remember responses for LWF items than HWF items (e.g.,

| Joordens & Hockley,2000).

- Citations: Joordens & Hockley (2000)

Missing or Unclear (Non-Statistical)

- Table 1 referenced but not provided in the text

Citations Referenced

- **Joordens & Hockley (2000)** -- WF mirror effect
- **Brainerd et al. (2006)** -- recollection rejection
- **Gallo (2004)** -- recollection rejection
- **Light (2012)** -- recollection rejection
- **Schmid et al. (2010)** -- recollection rejection

Experiment 2 -- Introduction

Section: exp_2

Rationale for This Experiment

The experiment was run to elucidate whether similar mirror effects would be obtained for items varying in word frequency when controlling for context variability, and to determine if the perceived memorability of low-frequency items is confounded by their contextual variability.

Relation to Prior Experiments

This experiment builds on Experiment 1 by removing the Guess and Recall-to-Reject response options (due to low usage and lack of differences in Experiment 1) and by extending the design to cross word frequency with context variability.

Theoretical Refinements (relative to paper-level intro)

independent mirror effects of WF and CV

| Previous studies examining WF when controlling for CV have shown that these variables produce independent mirror effects (Cook et al.,2006; Steyvers & Malmberg,2003).

- Citations: Cook et al. (2006); Steyvers & Malmberg (2003)

remember responses for low WF and low CV

both low WF and low CV words are independently given more Remember responses than high WF and high CV words (Cook et al.,2006).

- Citations: Cook et al. (2006)

New Elements Introduced

- Crossed word frequency (WF) with context variability (CV) to create four stimulus classes
- Removed the Guess and Recall-to-Reject response options

Constraints Addressed

- Controls for context variability to rule out confound between WF memorability and CV
- Addresses low usage of Guess and Recall-to-Reject options observed in Experiment 1

Missing or Unclear

- Specific predictions for Experiment 2 are not explicitly stated in the introduction

Citations Referenced

- **Steyvers & Malmberg (2003)** -- measure of WF and CV created from TASA corpus
- **Landauer, Foltz, & Laham (1998)** -- details of TASA corpus
- **Cook et al. (2006)** -- independent mirror effects of WF and CV; Remember responses

Experiment 2 -- Methods

Section: exp_2

Participants

- **N:** 34
- **Recruitment:** University of Georgia
- **Compensation:** course credit
- **Testing context:** tested individually in a session that lasted approximately 30 min
- **Exclusions:**
 - one participant excluded due to a software error

Design

Design statement: 2 (word frequency: low vs high) x 2 (context variability: low vs high) within-subjects design.

Predictors:

- **word frequency**
 - Levels: low, high
 - Manipulation: within-subjects
 - Factor type: nominal
- **context variability**
 - Levels: low, high
 - Manipulation: within-subjects
 - Factor type: nominal

Counterbalancing: not reported

Outcome Measures

None extracted

Materials

- **Description:** words grouped into four classes: LWF/LCV, LWF/HCV, HWF/LCV, HWF/HCV
- **Source:** adapted from Steyvers and Malmberg (2003)
- **Stimulus pool size:** 268
- **Stimuli presented per participant:** 80
- **Additional details:**
 - six taboo items removed from each class
 - 20 words randomly chosen from each class for study phase

Procedure

Ordered steps:

1. Study phase presenting 80 words (20 per class)
2. Test phase presenting studied words plus 80 new items (20 per class) randomly chosen

- **Apparatus:** computer software (unspecified)
- **Experimenter presence:** not reported
- **Presentation format:** random selection of words for study and test phases
- **Stimulus timing:** not reported
- **Session duration:** approximately 30 min

Preregistration and Analysis Plan

- **Preregistration:** not reported

Author-Provided Rationale for Methodological Choices

- Increased sample size from Experiment 1 due to the smaller number of items per class

Missing or Unclear

- no power analysis reported
- exact hardware/software details not specified
- stimulus timing parameters not reported
- full description of outcome measures missing
- counterbalancing scheme not described
- detailed procedural steps beyond study and test phases not provided
- experimenter presence not indicated

Citations Referenced in This Section

- **Steyvers and Malmberg (2003)** -- source of word corpus used for stimuli

Quoted Evidence from Source

Thirty-five undergraduate students from the University of Georgia participated in exchange for course credit.

We selected items from the Steyvers and Malmberg (2003) corpus.

One participant was excluded due to a software error.

Experiment 2 -- Results and Discussion

Section: exp_2

Statistical Approach

We do not report the interaction effects in Experiment 2 and 3 because there were no statistically significant interactions. One interaction was marginally significant in Experiment 3 and is thus reported.

Key Findings

Main effects of WF and CV on hit rates

- **Test:** 2×2 repeated measures ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, hit rates

Verbatim statistical report:

| $F(1,33) = 13.460, p = .001, \eta p^2 = .290$, and $F(1,33) = 19.763, p < .001, \eta p^2 = .375$

Main effects of WF and CV on Remember responses

- **Test:** 2×2 repeated measures ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Remember responses

Verbatim statistical report:

| $F(1,33) = 22.360, p < .001, \eta p^2 = .403$, and $F(1,33) = 39.710, p < .001, \eta p^2 = .546$

Main effects of WF and CV on Know responses

- **Test:** 2×2 repeated measures ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Know responses

Verbatim statistical report:

| $F(1,33) = 4.146, p = .050, \eta p^2 = .112$, and $F(1,33) = 4.878, p = .034, \eta p^2 = .129$

Main effect of WF and marginal effect of CV on correct rejection rates

- **Test:** 2×2 repeated measures ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, correct rejection rates

Verbatim statistical report:

| $F(1,33) = 22.527, p < .001, \eta^2 = .406$, and $F(1,33) = 3.873, p = .058$

Main effects of WF and CV on Would Have Remembered responses

- **Test:** 2×2 repeated measures ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Would Have Remembered responses

Verbatim statistical report:

| $F(1,33) = 15.808, p < .001, \eta^2 = .324$, and $F(1,33) = 57.276, p < .001, \eta^2 = .634$

Main effect of CV on Lacks Familiarity responses

- **Test:** 2×2 repeated measures ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Lacks Familiarity responses

Verbatim statistical report:

| $F(1,33) = 9.059, p = .005, \eta^2 = .205$, and $F(1,33) = .022, p = .833$

Main effects of WF and CV on conditionalized Would Have Remembered responses

- **Test:** 2×2 repeated measures ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, conditionalized Would Have Remembered responses

Verbatim statistical report:

| $F(1,33) = 10.138, p = .003, \eta^2 = .235$, and $F(1,33) = 34.196, p < .001, \eta^2 = .509$

Authors' Interpretations

Hit rates for LWF and LCV items

These main effects demonstrate higher hit rates for both LWF and LCV items.

Remember responses for LWF and LCV items

These results indicate more Remember responses for LWF and LCV items than HWF and HCV items.

Know responses for HWF and HCV items

The amount of Know responses was higher for the HWF and HCV items than the LWF and LCV items.

Would Have Remembered responses for LWF and LCV items

Thus, the increased perception of memorability for WF is not confounded with CV and vice versa.

Lacks Familiarity responses for HCV items

There were more Lacks Familiarity responses for HCV items than LCV items.

Similar but independent patterns of recognition decisions

In general, we found CV and WF produced similar, but independent patterns of recognition decisions, including subjective classifications.

Authors' Stated Links to Their Own Predictions

Replication of Cook et al. (2006) and Steyvers and Malmberg (2003)

We replicated Cook et al. (2006) and Steyvers and Malmberg (2003) in finding mirror effects for both WF and CV separately (marginally in the case of rejection rates for CV).

- Citations: Cook et al. (2006); Steyvers and Malmberg (2003)

Missing or Unclear (Non-Statistical)

- Descriptive statistics for individual conditions in Table 2 are not explicitly linked to the statistical tests in the text.
- Sample sizes for subgroups are not explicitly stated.
- Reliability estimates for measures are not reported.

Citations Referenced

- **Cook et al. (2006)** -- replication of mirror effects
- **Steyvers and Malmberg (2003)** -- replication of mirror effects

Experiment 3 -- Introduction

Section: exp_3

Rationale for This Experiment

The experiment was run to determine whether the differences in rejection classifications observed previously would still occur when word frequency (WF) is manipulated between subjects, using pure lists to avoid exposure to contrasting WF items.

Relation to Prior Experiments

This experiment extends Experiment 2 by keeping context variability (CV) manipulation within-subjects but changing the WF manipulation to a between-subjects design, thereby testing the effect of pure-list WF manipulations.

Specific Predictions

no difference in WHR responses by word frequency

If this is the case, there should be no differences in Would Have Remembered responses between LWF and HWF words.

Theoretical Refinements (relative to paper-level intro)

memorability sensitivity in pure lists

It is possible, thus, that participants are not sensitive to memorability differences in new items when presented with all LWF or HWF items because relative differences between the memorability for these items is not diagnostic of list membership.

New Elements Introduced

- WF manipulated between-subjects (pure list design)
- prevention of exposure to contrasting WF items
- CV still manipulated within-subjects

Constraints Addressed

- tests WF mirror effect in pure lists
- addresses prior failure to find correct rejection component in pure-list manipulations (Dewhurst et al., 2004)
- examines whether memorability differences are diagnostic when only one WF level is present

Missing or Unclear

- Explicit statement of expected effect of CV manipulation in this experiment
- Details on sample size or stimulus specifics

Citations Referenced

- **Clark & Burchett (1994)** -- WF mirror effect found in mixed and pure lists
- **Dorfman & Glanzer (1988)** -- WF mirror effect found in mixed and pure lists
- **Gorman (1961)** -- WF mirror effect found in mixed and pure lists
- **McCormack & Swenson (1972)** -- WF mirror effect found in mixed and pure lists
- **Dewhurst et al. (2004)** -- hit rate due to recollection; correct rejection component not found in pure list manipulation
- **Malmberg and Murnane (2002)** -- false alarm rate for LWF decreased with presence of more studied HWF items

Experiment 3 -- Methods

Section: exp_3

Participants

- **N:** 69
- **Recruitment:** University of Georgia
- **Compensation:** course credit
- **Testing context:** tested individually in a session that lasted approximately 30 min
- **Exclusions:**
- one participant excluded due to a software error

Design

Design statement: 2 (word frequency: low vs. high) between-subjects x 2 (consonant-vowel status: low vs. high) within-subjects mixed design.

Predictors:

- **word frequency condition**
 - Levels: low word frequency (LWF), high word frequency (HWF)
 - Manipulation: between-subjects
 - Factor type: nominal
- **consonant-vowel status**
 - Levels: low CV (LCV), high CV (HCV)
 - Manipulation: within-subjects
 - Factor type: nominal

Counterbalancing: not reported

Outcome Measures

None extracted

Materials

- **Description:** items selected from the Steyvers and Malmberg (2003) corpus
- **Source:** adapted from Steyvers and Malmberg (2003)
- **Stimuli presented per participant:** 60 studied items per participant; 120 test items per participant

Procedure

Ordered steps:

1. Study phase: participants studied 30 items of each CV class (LCV and HCV) within their assigned word-frequency condition
 2. Test phase: participants were tested on the 60 studied items plus 60 new items (30 new per CV class)
- **Apparatus:** not reported
 - **Experimenter presence:** not reported

- **Presentation format:** not reported
- **Stimulus timing:** not reported
- **Session duration:** approximately 30 min

Preregistration and Analysis Plan

- **Preregistration:** not reported

Missing or Unclear

- power analysis not reported
- stimulus timing details not reported
- apparatus (hardware/software) not reported
- verbatim participant instructions not provided
- counterbalancing scheme not described
- outcome measure definition not described
- total number of stimuli in the pool not specified
- detailed demographics (age, gender breakdown) not provided
- exact compensation amount not specified
- additional exclusion criteria not reported
- counterbalancing of study/test order not reported

Citations Referenced in This Section

- **Steyvers and Malmberg (2003)** -- source of stimulus items

Quoted Evidence from Source

Seventy undergraduate students from the University of Georgia participated in exchange for course credit.

Thirty-five participants were randomly assigned to a LWF condition and the other 35 were assigned to a HWF condition.

One participant was excluded due to a software error.

Experiment 3 -- Results and Discussion

Statistical Approach

Analyses were conducted similarly to Experiment 2 with the exception that WF was compared between-subjects. In a 2 (WF) - 2 (CV) mixed ANOVA on hits, the main effect of CV was significant, but the main effect of WF was not. We analyzed the raw proportions of Remember and Know responses in two separate 2 (WF) - 2 (CV) mixed ANOVAs. For Would Have Remembered responses, we found a main effect of CV, but no main effect of WF. For Lacks Familiarity responses, we also obtained a main effect for CV, but no main effect of WF. In terms of conditionalized Would Have Remembered responses, there was a main effect of CV, but no main effect of WF.

Key Findings

Main effect of CV on hit rates

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, hit rates

Verbatim statistical report:

| $F(1,67) = 31.972, p < .001, \eta_p^2 = .323$

No main effect of WF on hit rates

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, hit rates

Verbatim statistical report:

| $F(1,67) = 1.173, p = .283$

Main effect of CV on Remember responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Remember responses

Verbatim statistical report:

█ $F(1,67) = 42.340, p < .001, \eta_p^2 = .387$

No main effect of WF on Remember responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Remember responses

Verbatim statistical report:

█ $F(1,67) = .140, p = .709$

Main effect of CV on Know responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Know responses

Verbatim statistical report:

█ $F(1,67) = 8.164, p = .006, \eta_p^2 = .109$

No main effect of WF on Know responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Know responses

Verbatim statistical report:

█ $F(1,67) = .283, p = .596$

Main effect of CV on correct rejection rates

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, correct rejection rates

Verbatim statistical report:

█ $F(1,67) = 24.180, p < .001, \eta_p^2 = .265$

Main effect of WF on correct rejection rates

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, correct rejection rates

Verbatim statistical report:

| F (1,67) = 4.066, p = .048, η_p^2 = .057

Marginal interaction between WF and CV on correct rejection rates

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** interaction
- **Variables in model:** WF, CV, correct rejection rates

Verbatim statistical report:

| F (1,67) = 3.380, p = .070

Main effect of CV on Would Have Remembered responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Would Have Remembered responses

Verbatim statistical report:

| F (1,67) = 85.075, p < .001, η_p^2 = .559

No main effect of WF on Would Have Remembered responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Would Have Remembered responses

Verbatim statistical report:

| F (1,67) = .611, p = .437

Main effect of CV on Lacks Familiarity responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect

- **Variables in model:** WF, CV, Lacks Familiarity responses

Verbatim statistical report:

| $F(1,67) = 25.661, p < .001, \eta_p^2 = .277$

No main effect of WF on Lacks Familiarity responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, Lacks Familiarity responses

Verbatim statistical report:

| $F(1,67) = .422, p = .518$

Main effect of CV on conditionalized Would Have Remembered responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, conditionalized Would Have Remembered responses

Verbatim statistical report:

| $F(1,67) = 66.372, p < .001, \eta_p^2 = .498$

No main effect of WF on conditionalized Would Have Remembered responses

- **Test:** 2 (WF) - 2 (CV) mixed ANOVA
- **Effect type:** main effect
- **Variables in model:** WF, CV, conditionalized Would Have Remembered responses

Verbatim statistical report:

| $F(1,67) = .081, p = .776$

Authors' Interpretations

CV mirror effect

| We replicated the finding from Experiment 2 and previous studies (Cook et al.,2006; Steyvers & Malmberg,2003) that demonstrates a mirror effect such that LCV items have higher hit rates and lower false alarm rates than HCV items. Here, the correct rejection portion of the CV mirror effect was significant and not marginally significant. In addition,

there were more Remember and Would Have Remembered responses and less Know and Lacks Familiarity responses for LCV items. The results further support the notion that the CV mirror effect is related to increased recollection for the studied items and an increased use of a memorability-based judgment for the new items.

- Citations: Cook et al. (2006); Steyvers & Malmberg (2003)

WF mirror effect

Contrasting with Experiments 1 and 2, we failed to find a statistical advantage in the hit rates for LWF items as compared with HWF items. In addition, there were no differences in Remember and Know responses between these two classes of items (the numerical difference between Remember responses for LWF and HWF items was nearly nonexistent). We did find, however, that correct rejection rates were higher for LWF items. Despite this advantage, we did not find any differences in Would Have Remembered or Lacks Familiarity responses between LWF and HWF items (there was only a very slight numerical difference). Thus, even though we did replicate the correct rejection portion of the WF mirror effect, participants no longer classified LWF items as being words they Would Have Remembered more often than HWF items. This finding suggests that the relative/contextual distinctiveness across items influences the report of a memorability-based assessment associated with the rejection of unstudied items.

Authors' Stated Links to Their Own Predictions

Power analysis

We had .54 power to detect medium-sized effects and .90 power to detect a large sized-effects (Faul, Erdfelder, Buchner, & Lang, 2009).

- Citations: Faul, Erdfelder, Buchner, & Lang (2009)

Anomalous or Unexpected Results (as characterized by authors)

- The marginal interaction between WF and CV on correct rejection rates

Limitations Stated by Authors

- The failure to find some of these effects may be partially due to the reduced power of using a between-subjects as compared with a within-subjects design (though we will argue this is not the primary determinant)

Preview of Next Experiment

As will be explained more in Section 8, the failure to find some of these effects may be partially due to the reduced power of using a between-subjects as compared with a within-subjects design (though we will argue this is not the primary determinant).

Missing or Unclear (Non-Statistical)

- Descriptive statistics for individual conditions in Table 3 are not fully reported (some standard errors are missing)

Citations Referenced

- **Cook et al. (2006)** -- previous studies
- **Steyvers & Malmberg (2003)** -- previous studies
- **Faul, Erdfelder, Buchner, & Lang (2009)** -- power analysis

General Discussion

Section: paper-level

Summary of Findings

The study replicated correct rejection advantages for low-frequency words and showed that these advantages are driven by participants' perceived memorability rather than lack of familiarity or recollection, especially in mixed-list contexts; hit-rate advantages were not statistically evident in pure-list designs, and subjective classification patterns varied with list composition.

Theoretical Conclusions

memorability-driven rejection

participants capitalize on additional processes to classify lures in a recognition memory test as new. These processes are related to general memory mechanisms that support successful recognition (i.e., familiarity and perceived memorability).

inherent distinctiveness account

The infrequent occurrence of LWF items makes them inherently more distinctive and more memorable than HWF words. This inherent or secondary distinctiveness (Schmidt, 1991) likely accounts for the increased ability to reject these stimuli.

- Citations: Schmidt (1991)

contextual distinctiveness influence on judgments

When both LWF and HWF items are present, the expected average memorability would not be as high, thus resulting in more Would Have Remembered judgments for LWF items. This result suggests that a Would Have Remembered judgment is not just based on the innate characteristics of the item itself, but can also be influenced by the specific test context.

Cross-Experiment Integration

within- vs pure-list effects

In within-subjects (mixed-list) manipulations, we found mirror effects, but we also found that correct rejections to LWF items were more likely to be attributed to a memorability process... We also replicated the correct rejection advantage for LWF items in pure lists... the pattern of subjective classifications, however, did change in the pure-list manipulation.

Results Authors State Support Their Theory

memorability process advantage

correct rejections to LWF items were more likely to be attributed to a memorability process and not to a lack of perceived familiarity or a recollection rejection mechanism.

- Citations: Benjamin (2003); Brown et al. (1977); Guttentag & Carroll (1998)

Results Authors State Challenge a Theory

lack of hit rate advantage in pure lists

We did not find statistical evidence for the LWF hit rate advantage in pure-list manipulations, which is inconsistent with prior research.

- Citations: Dorfman & Glanzer (1988); Gorman (1961); McCormack & Swenson (1972)

Alternative Accounts Addressed by Authors

familiarity-based rejection (*named theory*)

- **Description:** Participants could reject LWF items based on an assessment that they do not have high levels of familiarity, thus not passing a criterion for having been studied.

- **How authors addressed it:** acknowledged and argued that even when afforded the Lacks Familiarity option, the rejection advantage was related to memorability rather than lack of familiarity.
- **Citations:** Arndt & Reder (2002); Guttentag & Carroll (1994); Joordens & Hockley (2000); Reder et al. (2000); Rugg et al. (1995)

distinctiveness heuristic (*named theory*)

- **Description:** Low-frequency items are inherently more distinctive, leading to higher memorability and easier rejection.
- **How authors addressed it:** proposed as an explanation for the increased rejection of LWF items, citing inherent distinctiveness.
- **Citations:** Schmidt (1991)

functional account of recollection (*named theory*)

- **Description:** Judgments of recollection depend on context and when Remember judgments are criterial for recognition decisions.
- **How authors addressed it:** drawn as an analogous explanation for rejection judgments, suggesting context may affect criterial status of Would Have Remembered responses.
- **Citations:** Bodner & Lindsay (2003); Gruppuso, Lindsay, & Kelley (1997)

Implications for the Field

subjective classifications utility

If used properly, these subjective classifications can be effectively utilized. Future research on these rejection classifications should parallel those that have been done with the Remember-Know paradigm...

- **Citations:** Donaldson (1996); Dunn (2004); Wais et al. (2008)

Limitations Stated by Authors

- requiring mutually-exclusive rejection responses may limit knowledge about the distribution of memorial characteristics
- assumption that participants equally weighted rejection options
- potential confound between confidence and process labels (e.g., Would Have Remembered may reflect confidence rather than distinct process)

Future Directions Proposed by Authors

- clarify additional processes beyond the three examined that may influence mirror effects
- differentiate more precisely between Would Have Remembered and Lacks Familiarity classifications
- use think-aloud protocols, confidence assessments, and neural measures to further examine rejection processes
- investigate how test context influences criterial status of rejection judgments

Citations Referenced

- **Glanzer et al. (1993)** -- mirror effects as tradeoff
- **Benjamin (2003)** -- LWF items more memorable
- **Brown et al. (1977)** -- LWF items more memorable
- **Guttentag & Carroll (1998)** -- LWF items more memorable
- **Schmidt (1991)** -- distinctiveness
- **Bodner & Lindsay (2003)** -- functional account
- **Gruppuso, Lindsay, & Kelley (1997)** -- functional account
- **McCabe & Balota (2007)** -- expectancy heuristic
- **Donaldson (1996)** -- subjective classifications
- **Dunn (2004)** -- subjective classifications
- **Wais et al. (2008)** -- subjective classifications
- **Migo et al. (2012)** -- Remember-Know paradigm
- **McCabe et al. (2011)** -- think aloud protocols
- **Lampinen et al. (2005)** -- think aloud protocols
- **Williams et al. (2013)** -- subjective classifications
- **Curran (2004)** -- neural correlates
- **Skinner & Fernandes (2007)** -- neural correlates
- **Spaniol et al. (2009)** -- neural correlates

Critique

Paper: Meeks, Knight, Brewer, Cook 2014 - Marsh Investigating the subjective reports of rejection processes in the word frequency mirror effect

Overall Assessment

The paper demonstrates moderate methodological and statistical shortcomings that undermine its scholarly rigor. While the experimental design addresses an interesting question, the lack of power analysis, incomplete reporting of ANOVA effects, and uncorrected multiple comparisons raise concerns about the robustness of the findings. Moreover, the theoretical framework is not tightly aligned with the evidence, as key constructs (e.g., distinctiveness) are inferred without direct measurement, and non-significant results are not adequately integrated. These issues collectively limit the paper's evidentiary support and overall contribution to the literature.

2. Evidence-Theory Alignment

- **The discussion generalizes the memorability-based rejection process to a distinctiveness account without directly measuring distinctiveness, overstating theoretical support.**
- Severity: **moderate**
- Justification: While the experiments provide evidence that participants report more "Would Have Remembered" responses for low-frequency items, they do not include any direct measure of item distinctiveness. The authors therefore extend their findings to a distinctiveness explanation, which is not empirically substantiated by the reported data. This overextension may mislead readers about the strength of support for the distinctiveness account.

Evidence:

- `general_discussion -- field theoretical_conclusions`: Authors claim "inherent distinctiveness" accounts for the effect, yet no distinctiveness metric was collected.
- `introduction -- field theoretical_framework`: Distinctiveness heuristic is presented as a related theory but not operationalized in the experiments.

The authors claim that memorability-driven rejection fully explains the correct rejection advantage, yet the data show non-significant differences in "Lacks Familiarity" and "Recall-to-Reject" responses, which are not integrated into the theoretical narrative.

- Severity: **moderate**
- Justification: The theoretical claim that memorability is the primary driver of the rejection advantage is not fully supported because alternative processes were examined and found null, yet the discussion minimizes these results. Proper alignment would require addressing how these null findings constrain the memorability account.

- Evidence:
- `results_and_discussion/exp_1` -- field `key_findings`: Non-significant effects for Lacks Familiarity ($p = .238$) and Recall-to-Reject ($p = .125$) are reported but downplayed in the interpretation.
- `general_discussion` -- field `results_stated_to_support_theory`: Authors emphasize memorability advantage while omitting discussion of the null findings for alternative processes.

5. Statistical Completeness

- **The 2 (WF) $\times$ 3 (Response Type) repeated measures ANOVA in Experiment 1 does not report the main effect of Response Type, which is an expected component of the factorial analysis.**
- Severity: **moderate**
- Justification: In a factorial ANOVA, the full model includes main effects for each factor and their interaction. While the interaction is reported, the omission of the Response Type main effect leaves the statistical description incomplete. Reporting the main effect is standard practice to allow readers to assess whether the interaction is driven by overall differences across response types.
- Evidence:
- `methods/exp_1` -- field `design`: Design statement specifies a 1 (word frequency) within-subjects factor; the ANOVA reported combines WF with a 3-level Response Type factor.
- `results_and_discussion/exp_1` -- field `key_findings`: Finding reports an interaction ($F(2, 48) = 4.208, p = .021$) and simple effects for each response type, but no statement of the main effect of Response Type.

6. Methodological Concerns

- **No power analysis was reported for any of the three experiments despite modest sample sizes.**
- Severity: **moderate**
- Justification: Without a priori power calculations, it is unclear whether the participant numbers (25, 34, and 69) provide sufficient sensitivity to detect the hypothesized effects. This limits confidence in null or marginal findings, especially for interaction effects that were not reported.

Evidence:

- methods/exp_1 -- field power_analysis: null (no power analysis reported)
- methods/exp_2 -- field power_analysis: null (no power analysis reported)
- methods/exp_3 -- field power_analysis: null (no power analysis reported)

Counterbalancing procedures for within-subjects manipulations are not described in any experiment.

- Severity: **moderate**
- Justification: Proper counterbalancing is essential to control order effects in repeated-measures designs. The absence of any description raises concerns about potential confounds influencing the observed word-frequency and context-variability effects.

Evidence:

- methods/exp_1 -- field counterbalancing: "not reported" despite a within-subjects word-frequency design
- methods/exp_2 -- field counterbalancing: "not reported" for the 2 × 2 within-subjects design
- methods/exp_3 -- field counterbalancing: "not reported" for the mixed design

Exclusion criteria are either absent or insufficiently specified across experiments.

- Severity: **moderate**
- Justification: Transparent exclusion criteria are needed to assess potential biases from data cleaning. The limited reporting makes it impossible to evaluate whether participant removal could have systematically affected the results.

Evidence:

- methods/exp_1 -- field exclusions: empty array; no criteria for data cleaning or participant removal
- methods/exp_2 -- field exclusions: only mentions one participant excluded due to a software error; no broader criteria
- methods/exp_3 -- field exclusions: one participant excluded for software error; no other exclusion rules

The study does not mention any preregistration or open-science practices.

- Severity: **minor**

- Justification: Preregistration helps guard against undisclosed analytic flexibility. Its absence reduces confidence that the reported analyses were pre-specified rather than exploratory.

Evidence:

- methods/exp_1 -- field preregistration_status: "not reported"
- methods/exp_2 -- field preregistration_status: "not reported"
- methods/exp_3 -- field preregistration_status: "not reported"

Reliability or internal consistency estimates for the subjective classification measures (e.g., "Would Have Remembered", "Lacks Familiarity") are not provided.

- Severity: **minor**
- Justification: Subjective response categories are central to the theoretical claims. Without reliability evidence, it is unclear whether these measures consistently capture the intended constructs across participants.
- Evidence:
 - methods/exp_1 -- field outcome_measures: multiple subjective response categories listed, but no reliability statistics reported
 - results_and_discussion/exp_2 -- field missing_or_unclear: "Reliability estimates for measures are not reported"
 - results_and_discussion/exp_3 -- field missing_or_unclear: "Reliability estimates for measures are not reported"

7. Statistical Robustness

- **Multiple pairwise t-tests were conducted without correction for family-wise error rate, increasing the risk of Type I errors.**
- Severity: **moderate**
- Justification: The authors test many related hypotheses across response types. Without adjusting alpha, the nominal 5 % threshold is inflated, so some reported significant effects (e.g., $p = .021$ for the interaction, $p = .015$ for Remember responses) may be false positives. A correction or multivariate approach would better control error rates.

Evidence:

- results_and_discussion/exp_1 -- field key_findings: Eleven independent t-tests (e.g., hit rate, Remember, Guess, Know, correct rejection, Would Have Remembered,

Lacks Familiarity, Recall to Reject, and several conditionalized measures) are reported with p-values but no mention of Bonferroni, Holm, or other correction.

A marginal effect ($p = .058$) for context variability on correct rejection rates in Experiment 2 is described as a "marginal effect" yet is used to support the claim of independent mirror effects.

- Severity: **minor**
- Justification: Treating a p-value just above .05 as evidence can overstate the robustness of the effect. The authors should qualify conclusions more cautiously or provide confidence intervals to clarify the precision of the estimate.

Evidence:

- `results_and_discussion/exp_2` -- field `key_findings`: The report states "Main effect of WF and marginal effect of CV on correct rejection rates" with $F(1,33)=3.873$, $p = .058$.

Effect sizes are reported for some significant tests but omitted for non-significant comparisons, limiting assessment of practical significance.

- Severity: **minor**
- Justification: Providing effect sizes for all comparisons, including null results, would allow readers to evaluate whether non-significant findings reflect negligible effects or simply insufficient power.
- Evidence:
 - `results_and_discussion/exp_1` -- field `key_findings`: Significant t-tests include Cohen's d (e.g., $d = .472$, $.526$), whereas non-significant tests (e.g., Know responses, Lacks Familiarity) report only p-values without effect size metrics.